Adhyyan Narang

MACHINE LEARNING RESEARCHER · **Ph.D., University of Washington**

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Research Profile

I study how feedback and optimization—from user feedback to fine-tuning data—shape LLM behavior, and how they can produce manipulation, deception, or other hidden misalignment. I combine empirical LLM research with reinforcement learning, game theory, and optimization to characterize these failures and develop mitigations.

Education

University of Washington **Seattle, WA**
Ph.D., Electrical and Computer Engineering; GPA: 3.96 *Sep. 2020–Mar. 2026*
Advisors: Maryam Fazel and Lillian Ratliff; Amazon Science Hub Fellow, 2023–2024

UC Berkeley **Berkeley, CA**
M.S., Electrical Engineering and Computer Sciences; GPA: 4.00 *2020*
Advisor: Anant Sahai

UC Berkeley **Berkeley, CA**
B.S., Electrical Engineering and Computer Sciences; Minor in Theater, Dance & Performance Studies *2019*
Overall GPA: 3.95; EECS GPA: 3.97; High Distinction

Research Experience

Independent Research **Seattle, WA**
Researcher, AI Safety *Apr. 2026–Present*

- Developed inference-time consensus methods that combine multiple fine-tuned LLMs to suppress hidden harmful behaviors without knowing which training sources were compromised.
- Designed and ran evaluations across explicit data poisoning and subtler fine-tuning failures; the methods reduced harmful behavior while preserving intended capabilities. Led writing and project coordination.

University of Washington **Seattle, WA**
Research Assistant *Sep. 2020–Mar. 2026*

- Led a project on how user choice shapes competing models, showing that feedback loops can drive overspecialization and developing a method to diagnose it; accepted to UAI 2026.
- Built and owned the evaluation pipeline for an ICLR 2025 study of targeted LLM manipulation, testing whether harmful behavior generalized across users and settings; implemented the experiments, analyzed results, and wrote the evaluation section.
- Developed theory and algorithms for sample-efficient reinforcement learning, learning in feedback loops, and interactive optimization.

BLISS Lab, University of California, Berkeley **Berkeley, CA**
Research Assistant *May 2019–Aug. 2020*

- Studied why classification can generalize better than regression in overparameterized models and why such models remain vulnerable to adversarial perturbations.

Publications

* *Equal contribution. Links are included in publication titles.*

Adhyyan Narang*, Artin Tajdini*, Claire Zhang, Jamie Morgenstern. "Inference-Time Consensus for Mitigating Hidden Behaviors from LLM Fine-Tuning." *Preprint, 2026.* **Accepted to the AdvML-Frontiers × CoTMA Workshop at COLM 2026.** *Code*

Adhyyan Narang, Sarah Dean, Lillian Ratliff, Maryam Fazel. "Dynamics of Learning under User Choice: Overspecialization and Peer-Model Probing." *UAI 2026*

Marcus Williams*, Micah Carroll*, **Adhyyan Narang**, Constantin Weisser, Brendan Murphy, Anca Dragan. "On Targeted Manipulation and Deception when Optimizing LLMs for User Feedback." *ICLR 2025.* **Press: New York Times, Washington Post. Workshop orals:** SATA and SoLaR

Publications (continued)

Adhyyan Narang, Andrew Wagenmaker, Lillian J. Ratliff, Kevin Jamieson. "Sample Complexity Reduction via Policy Difference Estimation in Tabular Reinforcement Learning." *NeurIPS 2024* **(Spotlight)**

Adhyyan Narang, Omid Sadeghi, Lillian J. Ratliff, Maryam Fazel, Jeff Bilmes. "Efficient Interactive Maximization of BP and Weakly Submodular Objectives." *UAI 2024*

Adhyyan Narang, Evan Faulkner, Dmitriy Drusvyatskiy, Maryam Fazel, Lillian J. Ratliff. "Multiplayer Performative Prediction: Learning in Decision-Dependent Games." *JMLR, 2023*

Adhyyan Narang, Evan Faulkner, Dmitriy Drusvyatskiy, Maryam Fazel, Lillian J. Ratliff. "Learning in Stochastic Monotone Games with Decision-Dependent Data." *AISTATS 2022*

Yue Sun, **Adhyyan Narang**, Halil Ibrahim Gulluk, Samet Oymak, Maryam Fazel. "Towards Sample-Efficient Overparameterized Meta-Learning." *NeurIPS 2021*

Tanner Fiez, Lillian J. Ratliff, Eric Mazumdar, Evan Faulkner, **Adhyyan Narang**. "Global Convergence to Local Minmax Equilibrium in Classes of Nonconvex Zero-Sum Games." *NeurIPS 2021*

Vidya Muthukumar*, **Adhyyan Narang***, Vignesh Subramanian*, Mikhail Belkin, Daniel Hsu, Anant Sahai. "Classification vs. Regression in Overparameterized Regimes: Does the Loss Function Matter?." *JMLR, 2021*

Adhyyan Narang, Vidya Muthukumar, Anant Sahai. "Classification and Adversarial Examples in an Overparameterized Linear Model: A Signal Processing Perspective." *ICML Workshop on Overparameterization, 2021*

Industry Experience

Amazon Robotics **Seattle, WA**
Research Scientist Intern *Jun. 2023–Sep. 2023*

- Used contextual bandits to optimize warehouse-robot item placement, increasing the organized-stow rate from 55% to 70%.

Amazon Customer Service AI **Seattle, WA**
Research Scientist Intern *Jun. 2022–Sep. 2022*

- Built a proof-of-concept customer-service dialogue system using inverse reinforcement learning and the MultiWOZ dataset.

Uber Eats **San Francisco, CA**
Machine Learning Engineering Intern *May 2018–Aug. 2018*

- Built a stochastic-optimization system for personalized promotions and deployed it to more than 20,000 users.

Veritas Technologies **Mountain View, CA**
Data Engineering Intern *Jun. 2017–Aug. 2017*

- Built an Apache Spark service that automated ML pipelines and reduced project incubation time by 30–40%.

Teaching

University of Washington, Teaching Assistant. Foundations of Dynamical Systems (Winter 2026); Foundations of Optimization and Machine Learning (Fall 2025, Spring 2022); Introduction to Probability (Spring 2025).

UC Berkeley, Head Content Teaching Assistant. Led homework and exam design for EE 16A, a 700-student course (Spring 2020).

Awards & Honors

Amazon Science Hub Fellow, 2023–2024; B.S. with High Distinction, UC Berkeley; Phi Beta Kappa; Tau Beta Pi; Dean's List (all undergraduate semesters).

Technical Skills

Python; PyTorch; Hugging Face Transformers; NumPy/SciPy; Weights & Biases; LoRA and LLM fine-tuning; reinforcement learning; experiment design and evaluation.